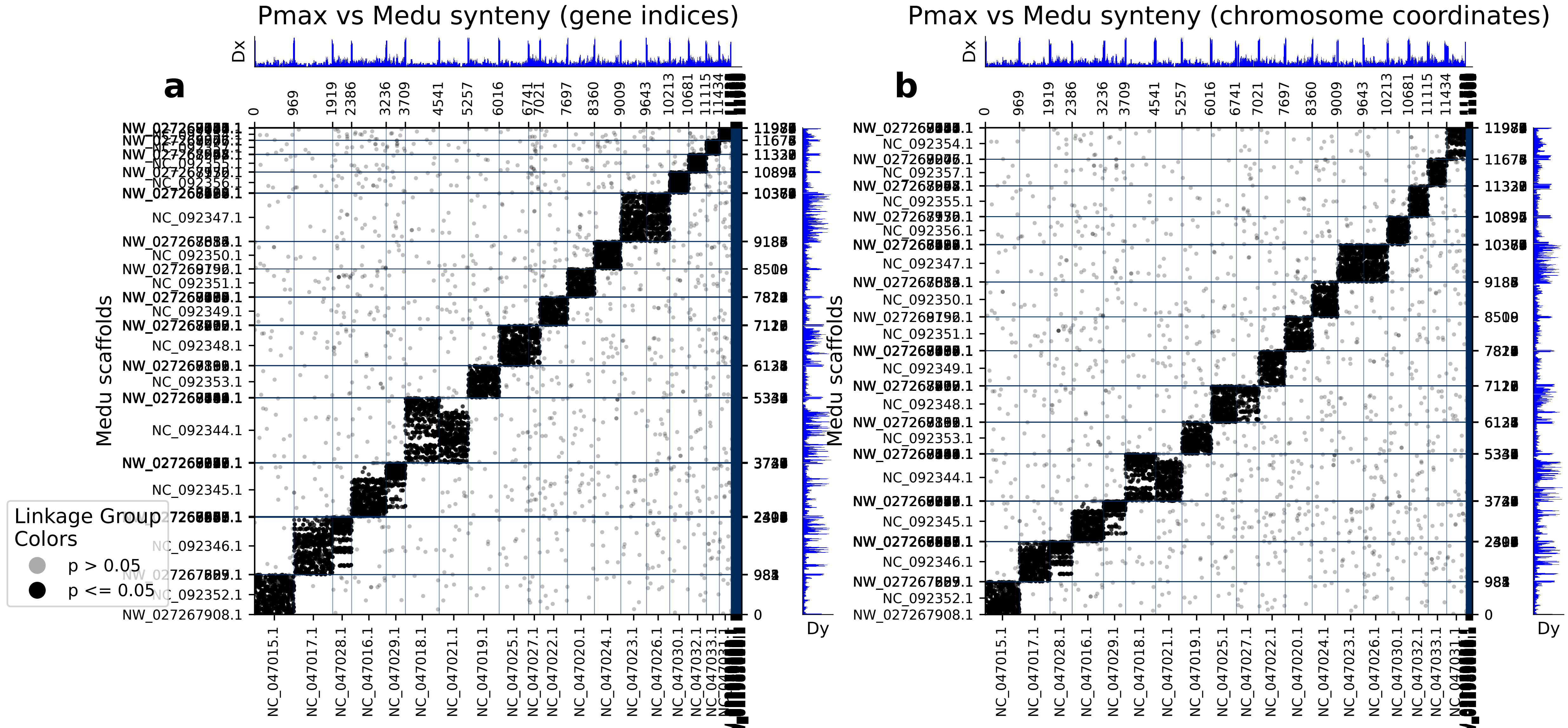
diamond: 11988 orthologs on 256 Pmax scaffolds and 137 Medu scaffolds.



c

Pmax vs Medu groups

Pmax scaffolds

d

Pmax vs Medu FET, whole chroms.

	Pmax scaf	Medu scaf	p (FET)	Count
0	NC_047015.1	NC_092352.1	0.00e+00	896
1	NC_047017.1	NC_092346.1	0.00e+00	889
2	NC_047018.1	NC_092344.1	0.00e+00	784
3	NC_047016.1	NC_092345.1	0.00e+00	728
4	NC_047019.1	NC_092353.1	0.00e+00	689
5	NC_047021.1	NC_092344.1	0.00e+00	654
6	NC_047025.1	NC_092348.1	0.00e+00	650
7	NC_047020.1	NC_092351.1	0.00e+00	580
8	NC_047022.1	NC_092349.1	0.00e+00	579
9	NC_047023.1	NC_092347.1	0.00e+00	577
10	NC_047024.1	NC_092350.1	0.00e+00	562
11	NC_047026.1	NC_092347.1	0.00e+00	485
12	NC_047030.1	NC_092356.1	0.00e+00	407
13	NC_047029.1	NC_092345.1	0.00e+00	406
14	NC_047028.1	NC_092346.1	1.54e-290	387
15	NC_047032.1	NC_092355.1	0.00e+00	350
16	NC_047033.1	NC_092357.1	0.00e+00	256
17	NC_047031.1	NC_092354.1	0.00e+00	216
18	NC_047027.1	NC_092348.1	8.08e-158	203
19	NW_022980858.1	NW_027267041.1	4.88e-04	2

e

Pmax vs Medu FET, chrom. pieces

Pmax scaffolds

	Pmax scaf	Medu scaf	p (FET)	Count
0	NC_047015.1 : 0-end	NC_092352.1 : 0-end	0.00e+00	896
1	NC_047017.1 : 0-end	NC_092346.1 : 0-end	0.00e+00	889
2	NC_047018.1 : 0-end	NC_092344.1 : 0-end	0.00e+00	784
3	NC_047016.1 : 0-end	NC_092345.1 : 0-end	0.00e+00	728
4	NC_047019.1 : 0-end	NC_092353.1 : 0-end	0.00e+00	689
5	NC_047021.1 : 0-end	NC_092344.1 : 0-end	0.00e+00	654
6	NC_047025.1 : 0-end	NC_092348.1 : 0-end	0.00e+00	650
7	NC_047020.1 : 0-end	NC_092351.1 : 0-end	0.00e+00	580
8	NC_047022.1 : 0-end	NC_092349.1 : 0-end	0.00e+00	579
9	NC_047023.1 : 0-end	NC_092347.1 : 0-end	0.00e+00	577
10	NC_047024.1 : 0-end	NC_092350.1 : 0-end	0.00e+00	562
11	NC_047026.1 : 0-end	NC_092347.1 : 0-end	0.00e+00	485
12	NC_047030.1 : 0-end	NC_092356.1 : 0-end	0.00e+00	407
13	NC_047029.1 : 0-end	NC_092345.1 : 0-end	0.00e+00	406
14	NC_047028.1 : 0-end	NC_092346.1 : 0-end	1.54e-290	387
15	NC_047032.1 : 0-end	NC_092355.1 : 0-end	0.00e+00	350
16	NC_047033.1 : 0-end	NC_092357.1 : 0-end	0.00e+00	256
17	NC_047031.1 : 0-end	NC_092354.1 : 0-end	0.00e+00	216
18	NC_047027.1 : 0-end	NC_092348.1 : 0-end	8.08e-158	203
19	NW_022980858.1 : 0-end	NW_027267041.1 : 0-end	4.88e-04	2

(a) depicts the chromosomes/scaffolds of Pmax (x-axis) plotted against the chromosomes/scaffolds of Medu (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11988 orthologs found between 256 Pmax scaffolds and 137 Medu scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.